Symlit_ARROWS_Study1

OSF: https://osf.io/2yudb

Internet_Archive: https://archive.org/details/osf-registrations-2yudb-v1

1) Data collection. Have any data been collected for this study already?

Yes, data collection is almost complete. Please see question 8.

2) Hypothesis. What's the main question being asked or hypothesis being tested in this study?

We investigate young children's comprehension of deictic reference in the graphic domain. Two-, three-, and four-year-old children are presented with an object-choice-task in which a cartoon agent is providing cues to help children choose between two options presented at the left and right bottom of the screen. The cues are either triangle cues placed in a central position between the options and pointing to one of them (*arrow condition*), or circles that are placed directly on the choice options (*marker condition*).

We code as correct when children choose the option that the triangle is pointing to or that the marker is placed on. We are using age as continuous predictor in a within-subjects design (*arrow*, *marker*) and investigate when children's group performance surpasses chance level in either condition. We hypothesize that (1) children will succeed earlier in the marker condition than in the arrow condition. We expect (2) both groups to perform above chance in the fourth year.

3) Dependent variable. Describe the key dependent variable(s) specifying how they will be measured.

The key dependent variable is children's choice behavior in the object-choice-task. Choice items are presented on the left and right side of the screen. Children respond by pointing to the side of their choice. Children's choices are coded as correct (1) or incorrect (0). We code as correct when children choose the option that the triangle is pointing to or that the marker is placed on.

4) Conditions. How many and which conditions will participants be assigned to?

In a within-subjects design, children will be presented with an equal number of trials of the arrow and marker condition.

5) Analyses. Specify exactly which analyses you will conduct to examine the main question/hypothesis.

All analyses will be computed in R (R Core Team, 2018). Bayesian models will be run in Stan (http://mc-stan.org/) and implemented via the function brm of the package brms (Bürkner, 2017). We will use logistic Bayesian generalized linear mixed models (GLMM) to fit children's responses (0/1) as a function of their absolute age in days, task (arrow, marker) and an interaction between trial and task. We use default priors and include trial and sex as fixed effects to be controlled for. Trial number will be added as a random slope within subject.

Model: correct ~ task*z.age +z.trial +z.sex +(z.trial|id)

- correct: correct choice (0/1);
- task: arrow condition or marker condition;
- z.age: age in days, centered to a mean of 0
- z.trial: trial number, scaled
- z.sex: participants' sex (male/female), scaled
- id: participant id

The analysis will model participants binary choices to predict the probability of children interpreting the cues correctly and model how this probability will change as a function of their absolute age in days. In order to evaluate the relevance of age and task type for children's performance, we will compare a full model as specified above with a reduced model lacking the interaction of age and task by using WAIC scores and weights (McElreath, 2016). Furthermore, we will inspect the model estimates for the different predictors (including their 95% Credible Interval (CrI)). In addition, we use the model to predict the developmental trajectory (with 95% CrI) for each task type. The criterion for settling when children perform above chance with either type of stimuli is the point at which the 95% CrI for a particular trajectory does no longer overlap with a midline demarcating the 50% chance level.

To further explore our data, we also bin participants by year of age. To test whether group-level performance is above chance in all age groups, we use two-tailed one-sample t-tests with the chance level set to .5. We provide Cohen's d as a standardized effect size for significance testing (computed via the function `cohensD`). Additionally, each participant's data will also be submitted to a binomial test to determine whether their performance is above chance on an individual level.

6) Outliers and Exclusions. Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Prior to the main study phase, children are tested with familiarization trials in which they simply have to point to the position of an item that was hidden in plain sight. This serves to ensure that children are comfortable and understand the search game.

In order to enter the main study phase, children should have a success rate of 66% in the familiarization. The experimenter provides at least four trials. If three in four are correct (75%), the child proceeds to the main phase. If not, two additional familiarization trials are run. If the child is correct in four out of six trials (66%), she proceeds to the main study phase. If not, two additional trials are presented. If six in eight are correct (75%), the child enters the main study phase. If children do not fulfill the criterion in a maximum of eight trials, they are dropped from the sample.

Children will also be excluded from the sample if they do not provide at least eight valid out of 16 possible trials in the main study phase. Trials are excluded if children do not respond, if responses cannot be interpreted, if parents or caregivers interfere with the procedure, or on the basis of technical problems or experimenter error.

7) Sample Size. How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Starting at 24 months, we will test two children per month over an age-range of 36 months for a total sample of 72 children. Each child contributes a minimum of eight and a maximum of 16 trials. While using age as a continuous predictor is rare in developmental studies, we aim at having a sample of at least 24 children per year of age (two-, three-, and four-year-olds) which is in line with conventions in the field. In case children in either condition do not perform above chance as a group within the current age-range, the age-range will be extended by a year increasing the sample by 24 children to determine when performance surpasses chance-level.

8) Other. Anything else you would like to pre-register?

Data collection for the main sample began in July 2021 and is nearly complete by the time of preregistration. This preregistration mainly serves to commit to the analysis plan outlined above. However, this study is part of a three-year funded research project. The study design, hypotheses and analysis plan are in line with the research proposal submitted in February, 2019.

Before beginning data collection in July of 2021, we ran two pilot versions with small samples (N = 4) of two-, three- and four-year-old children to evaluate the age-appropriateness of the task and our stimulus design. We document all pilots and the specific conclusions we drew from them in the supplemental material accompanying the manuscript.

9) Name. Give a title for this AsPredicted pre-registration

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